Name _____ Student ID _____

Part I: Multiple Choice (20 points)

Circle the one correct answer in each question.

1. What is a trap in an operating system?

   (a) It transfers control to an OS handler; system calls may use this mechanism.
   (b) It lets a user process capture another process's output.
   (c) It is a scheduling policy that switches between ready processes.
   (d) It detects bugs without changing the CPU's execution mode.
   (e) Operating systems have no trap mechanism.

2. What protection risk arises if user mode and kernel mode are not separated?

   (a) Every process is automatically isolated from all other processes.
   (b) The operating system loses access to disks and external devices.
   (c) A process may access another process's memory.
   (d) Direct device access is automatically blocked by the operating system.
   (e) Creating child processes becomes impossible.

3. Which set gives the four components of a process in the model used in class?

   (a) Program and data only
   (b) Program, data, and control state
   (c) Program, data, and process state
   (d) Data, control state, and process state
   (e) Program, data, control state, and process state

4. Which OS goal is specifically intended to protect one application from another?

   (a) Performance
   (b) Isolation
   (c) Reliability
   (d) Portability
   (e) All four goals provide isolation.

5. Which mechanism lets the OS regain control from a CPU-bound process in a preemptive scheduler?

   (a) MMU
   (b) Timer interrupt
   (c) Cache
   (d) TLB
   (e) None of these.

6. Which part of a process is stored in kernel space?

   (a) Text segment
   (b) Data segment
   (c) Stack segment
   (d) Process descriptor
   (e) None; all four are stored in user space.

7. In a conventional process address space, which region grows in the opposite direction from the heap?

   (a) Code segment
   (b) Heap
   (c) Stack
   (d) Data segment
   (e) All regions grow in the same direction.

8. Can a process move directly from the ready state to the blocked state?

   (a) Yes, when it requests a file read.
   (b) Yes, when it reads a page from disk.
   (c) Yes, when a timer interrupt occurs.
   (d) No. It must first be running.
   (e) Yes, after either (a) or (b).

9. Which event can move a process into the ready state?

   (a) A timer interrupt preempts it.
   (b) Its I/O operation completes.
   (c) The process is created and admitted to the ready queue.
   (d) Any of the events in (a), (b), or (c).
   (e) None of the events in (a), (b), or (c).

10. What does it mean for a process to be a zombie?

    (a) It is running on the CPU and cannot be preempted.
    (b) It is waiting for I/O indefinitely.
    (c) It is ready but has never been selected to run.
    (d) It has terminated, but its parent has not yet collected its exit status.
    (e) Operating systems have no zombie state.

11. Which process-control block stores a process's state and saved CPU context?

    (a) PCB
    (b) TLB
    (c) Stack
    (d) MMU
    (e) Queue

12. Which situation can leave a zombie process in the system?

    (a) A child exits while its parent is alive and does not collect its exit status.
    (b) A parent creates several children and calls wait() for each one.
    (c) A child creates another process, and both parents collect their children's exit statuses.
    (d) A parent calls wait() and collects the child's exit status.
    (e) A child calls exec() and continues running the new program.

13. If a parent successfully calls wait() before its own print statement, what is guaranteed?

    (a) The child always prints after the parent.
    (b) A zombie process is always created.
    (c) The child starts again from the beginning of main().
    (d) A segmentation fault always occurs.
    (e) The parent's print statement runs only after the child terminates.

14. The following C program:

```
#include <stdio.h>
#include <unistd.h>

int main(void) {
    fork();
    fork();
    printf("hello");
    return 0;
}
```

    Assume both calls succeed. How many times will the word “hello” be printed?

    (a) Only once
    (b) Twice
    (c) Three times
    (d) Four times
    (e) Eight times

15. Which events can cause a running process to give up the CPU?

    (a) A timer interrupt preempts it.
    (b) It requests blocking I/O.
    (c) It calls wait() and must block.
    (d) It calls yield().
    (e) Any of the events in (a), (b), (c), or (d).

16. Which statement is NOT true about MLFQ?

    (a) It has multiple queues at different priority levels.
    (b) A new job enters at the lowest priority level.
    (c) A ready job is on exactly one queue.
    (d) Round robin is used among jobs in the same queue.
    (e) A periodic priority boost can move jobs to the highest-priority queue.

17. In the C storage-class model covered in class, which class requests register storage for a local variable?

    (a) Auto
    (b) Register
    (c) Extern
    (d) Static
    (e) None of these.

18. Which statement about partition allocation is true?

    (a) Single-partition allocation supports several processes in memory at once.
    (b) Flexible-size partitions eliminate external fragmentation.
    (c) Fixed-size partitions can suffer from internal fragmentation.
    (d) Compaction eliminates internal fragmentation in fixed-size partitions.
    (e) None of these statements is true.

19. What does the short-term process scheduler do?

    (a) Selects a program to load into memory.
    (b) Selects a ready process to run.
    (c) Selects a process to remove from memory.
    (d) Selects a memory region to assign to a process.
    (e) Selects a process to load into memory.

20. What does a page-replacement algorithm do when a frame is needed?

    (a) Selects the requested page to load into memory.
    (b) Selects a page to execute.
    (c) Selects a resident page to evict so its frame can be reused.
    (d) Selects a memory region to assign to a page.
    (e) Selects a process to remove from memory.

21. Which statement about the address-translation schemes covered in class is true?

    (a) A page table maps segment numbers to base-and-bounds pairs.
    (b) A segment table maps page numbers to physical frames.
    (c) A TLB stores page contents rather than address translations.
    (d) Every scheme requires both a page table and a segment table.
    (e) Paging uses page tables, segmentation uses segment tables, and a TLB may cache translations.

22. What is a disadvantage of segmentation when each segment occupies contiguous physical memory?

    (a) Address translation from virtual to physical addresses is impossible.
    (b) It cannot protect one process's memory from another process.
    (c) It can suffer from external fragmentation.
    (d) It always requires a large table for address translation.
    (e) It has no disadvantages.

23. What does demand paging mean?

    (a) A page is loaded only when it is referenced and is not already resident.
    (b) Every page is kept in physical memory for on-demand access.
    (c) All pages are loaded when a process starts.
    (d) Pages are loaded in advance based on predicted use.
    (e) Operating systems do not use demand paging.

24. Which step does NOT occur while handling a valid page fault?

    (a) The OS saves the faulting process's context.
    (b) If no free frame is available, the OS selects a victim page and reuses its frame.
    (c) The OS locates the requested page and reads it from disk.
    (d) The OS updates the page-table entry to mark the page present.
    (e) All of these steps may occur.

25. What does each segment-table entry contain in the scheme shown in Lecture 5?

    (a) A base-and-bounds pair.
    (b) A logical page number and a physical frame number.
    (c) A process's CPU register context.
    (d) A process ID and scheduling priority.
    (e) All of the items in (a)--(d).

26. With a fixed virtual address-space size, what is one effect of using a larger page size?

    (a) More external fragmentation.
    (b) Smaller TLB reach.
    (c) A smaller page table.
    (d) More page-table entries for each process.
    (e) Page size has no effect.

27. Which single-queue policy rotates among all ready jobs, giving each a time slice in turn?

    (a) FIFO
    (b) STCF
    (c) RR
    (d) MLFQ
    (e) SJF

28. Which nonpreemptive policy serves ready jobs in arrival order until each finishes or blocks?

    (a) FIFO
    (b) STCF
    (c) RR
    (d) MLFQ
    (e) None of these policies.

29. Where is a nonresident virtual-memory page typically kept?

    (a) In a TLB.
    (b) In a page-table entry.
    (c) In an available physical frame.
    (d) In the CPU cache.
    (e) On secondary storage.

30. What is the response time of a process?

    (a) Time from the process's first CPU allocation until completion.
    (b) Time from the process's arrival until completion.
    (c) Time from the process's arrival until it first runs.
    (d) Time from when the process runs until it requests I/O.
    (e) Time the process runs during one CPU time slice.

31. What is the turnaround time of a process?

    (a) Time from the process's first CPU allocation until completion.
    (b) Time from the process's arrival until completion.
    (c) Time from the process's arrival until it first runs.
    (d) Time from when the process runs until it requests I/O.
    (e) Time the process runs during one CPU time slice.

32. What is a common problem with variable-size (flexible) partitions?

    (a) They limit the degree of multiprogramming to one process.
    (b) They cause internal fragmentation between allocated partitions.
    (c) They make address translation impossible.
    (d) They can cause external fragmentation.
    (e) They have no fragmentation problems.

33. With a single-level page table and fixed page size, what is a cost of a larger virtual address space?

    (a) More memory is needed to store the page table.
    (b) Internal fragmentation is eliminated.
    (c) External fragmentation increases.
    (d) TLB reach becomes smaller.
    (e) A larger address space has no cost.

34. How can a multilevel page table save memory for a sparse address space?

    (a) It eliminates the need for a page table.
    (b) It reduces the size of the virtual address space.
    (c) It increases the page size.
    (d) It allocates every lower-level table, even for unused regions.
    (e) It can omit lower-level page tables for unused regions.

35. In the simplified state model, which state does a newly created process enter after admission?

    (a) Ready
    (b) Running
    (c) Blocked
    (d) Terminated
    (e) None of these.

36. Which state is a process in while it is executing on the CPU?

    (a) Ready
    (b) Running
    (c) Blocked
    (d) Terminated
    (e) New

37. What fairness risk can SJF create if shorter jobs keep arriving?

    (a) It guarantees the highest average waiting time.
    (b) It always increases average response time.
    (c) It prevents short jobs from running before long jobs.
    (d) It gives every job an equal share of CPU time.
    (e) A long job may starve.

38. Which scheduling metric is Round Robin primarily designed to improve?

    (a) Minimum average turnaround time.
    (b) Minimum average waiting time.
    (c) Zero starvation under every workload.
    (d) Lower average response time for interactive jobs.
    (e) Maximum throughput for every time quantum.

39. Which statement about paging is correct?

    (a) Pages and page frames may have different sizes.
    (b) Smaller pages always produce smaller page tables.
    (c) Transferring a larger page takes no more time than transferring a smaller page.
    (d) A page belongs to logical memory; a frame belongs to physical memory.
    (e) Demand paging lets a program request that all pages be preloaded.

40. Which page-replacement policy is an ideal benchmark but cannot be implemented exactly without knowing future references?

    (a) Optimal
    (b) FIFO
    (c) LRU
    (d) LFU
    (e) All of these policies can be implemented exactly.

41. When does a page fault occur?

    (a) A process references a valid virtual page whose present bit is 0.
    (b) A process accesses an address outside its virtual address space.
    (c) A process accesses a page that is already present in physical memory.
    (d) A process accesses a present page that has been modified.
    (e) All of the above cause a page fault.

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Part II: Short Answer (10 points)

Answer all questions briefly in the spaces provided. (1 point each.)

1. Write the full names of MLFQ and STCF.

2. What is the difference between a process and a program?

3. Expand MMU and state its address-translation function.

4. Give two examples of CPU state saved during a context switch.

5. Give two types of information stored in a PCB, other than CPU register values.

6. What is an exited child process called before its living parent reaps its exit status?

7. Give one example each of an I/O-intensive task and a CPU-intensive task.

8. What is thrashing in a virtual-memory system?

9. With base = 16 KiB and bound = 4 KiB, is virtual byte address 4200 valid? Why?

10. Expand TLB and explain why a TLB miss need not cause a page fault.

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Part III: Written Questions (20 points)

Answer all three questions in the answer booklet. Show your work.

1. **(8 points)** Consider the following set of processes (jobs), where the run time of the process is given in milliseconds (ms):

Process	Run Time	Priority
P_1	10	3
P_2	1	1
P_3	2	3
P_4	1	4
P_5	5	2

   The processes are assumed to arrive in the order P_1, P_2, P_3, P_4, P_5, all at time 0.

   (a) Draw three diagrams illustrating the execution of these processes using FIFO, SJF, and RR scheduling with time slice = 1 ms.

   (b) What is the Turnaround time of each process for each of the schedulers in part (a)?

   (c) What is the Response time of each process for each of the schedulers in part (a)?

   (d) Which of the schedulers results in (i) the minimum average Turnaround time, (ii) the minimum average Response time (taking the average over all processes)?

2. **(4 points)** Explain how a commercial bank handles customer during rush hours with the concept of the Multi-level Feedback Queue (MLFQ), in your opinion. Describe the algorithms and policies used and provide suggestions on how to improve the process with the goal of optimizing the average turnaround time of customers.

3. **(8 points)** Draw three tables show the 3 Frames in physical memory using FIFO, LRU and OPT for page replacement policy (separate table for both policies) and find the Effective Access Time (EAT) of each policies when references strings (list of needed pages) as below:

   7, 0, 1, 3, 0, 3, 2, 4, 2, 7, 0, 3, 2, 1, 2, 0, 1, 4, 0, 1 (20 needed pages)

   Assume that the memory access time is 10 ns and the disk access time is 10 ms.

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